Substitution

1 Find the value of:

a $9 + m$ when $m = 8$

b $14 + v$ when $v = 15$

c $n - 2$ when $n = 5$

d $n - 8$ when $n = 44$

e $\frac{n}{7}$ when $n = 14$

f $\frac{u}{9}$ when $u = 54$

g $u \times 5$ when $u = 6$

h $n \times 9$ when $n = 8$

i $2x + 9$ when $x = 4$

j y^2 when $y = -2$

k $x^2 - 4x$ when $x = 3$

l $\frac{x + 10}{4}$ when $x = 2$

m $\frac{3x}{4}$ when $x = 8$

n $2(x + 7)$ when $x = 4$

o $(x + 2)(x - 1)$ when $x = -3$

p $4y^2 + 9$ when $y = 2$

2 Find the value of:

a $m + n$ when $m = 7$ and $n = 12$

b $4x + 2y + 3$ when $x = 5$ and $y = 4$

c $x^2 - 4y$ when $x = -2$ and $y = 5$

d $8a + 4b - 9$ when $a = 2$ and $b = 5$

e $\frac{5xy}{9}$ when $x = 3$ and $y = 6$

f abc when $a = 3, b = -2$ and $c = -1$

3 For each formula find the value of y for the given value of x:

a $y = 6x - 9$, when $x = 12$.

b $y = \frac{x}{9} + 6$, when $x = 63$.

c $y = 4 - 3x$, when $x = 7$.

d $y = 14x + \frac{18}{5}$, when $x = \frac{1}{10}$.

Tables of values

4 Consider the following formulae:

a $q = p + 7$

i Complete the table:

p	1	2	3	4	5	6
q	8	9	10			

ii Find the value of q when $p = 30$.

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b $q = 3p$

i Complete the table:

p	1	2	3	4	5	6
q	3	6	9			

ii Find the value of q when $p = 20$.

c $q = 6p + 4$

i Complete the table:

p	1	2	3	4	5	6
q	10	16	22			

ii Find the value of q when $p = 40$.

d $n = m - 7$

iComplete the table:

m	10	11	12	13	14	15
n	3	4	5			

ii Find the value of n when $m = 20$.

e $q = 5p - 3$.

i Complete the table:

p	1	2	3	4	5	6
q	2	7	12			

ii Find the value of q when $p = 20$.

f $q = p^2 + 5$.

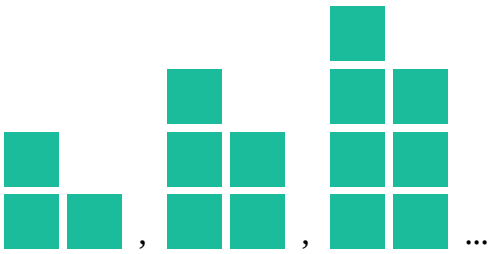
i Complete the table:

p	1	2	3	4	5	6
q	6	9	14			

ii Find the value of q when $p = 20$.

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5 Vanessa is making the sequence to the right out of tiles. She creates a table comparing the sequence number of a shape to the number of tiles needed to make it:

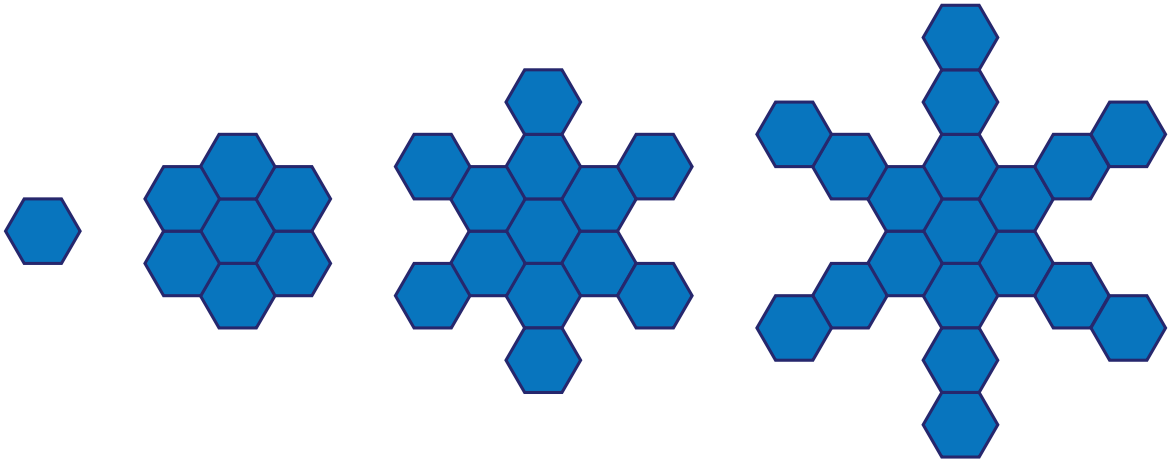


Sequence number (S)	1	2	3	4	5	6
No. tiles (T)	3	5	7			

a Write an algebraic rule for the number of tiles, T, in terms of the sequence number, S.

b Use your rule to complete the table of values.

6 James is making snowflakes out of hexagonal tiles:



James creates a table comparing the width of a snowflake to the number of tiles needed to make it:

Width (W)	1	3	5	7	9	11	13
No. tiles (T)	1	7	13	19			

a Write an algebraic rule for the number of tiles, T, in terms of the snowflake's width, W.

b Use your rule to complete the table of values.

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